

Priti Singh, PhD
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Location: Mumbai, India

Career Objective

Aspiring to pursue a dynamic career in developing advanced software solutions, computational models, and simulation frameworks, leveraging my expertise in C++, mathematical modeling, and numerical methods. I am keen on contributing to innovative projects that integrate software development with scientific and engineering principles, aiming to create impactful solutions in computational engineering and scientific modeling.

Education

- **PhD in Engineering Science**, Homi Bhabha National Institute (HBNI), Mumbai – BARC
Year of Passing: 2024 — 85% (in course work)
- **M.E. in Power Electronics**, RGPV, Bhopal – SGSITS, Indore
Year of Passing: 2015 — 88%
- **B.E. in Electrical and Electronics Engineering**, RGPV, Bhopal – IES IPS Academy, Indore
Year of Passing: 2010 — 74.59%

Key Skills

- C++ programming, High Performance and Parallel Computing (MPI coding), MATLAB programming, Simulink simulations, Computational Engineering, Mathematical modeling, Algorithm development, Finite element analysis, Computational Physics, Computational fluid dynamics, Qt, Numerical Methods, Optimization methods, Plasma simulation, Simulation and Modeling, Electromagnetics, Power system optimization.

Professional Experience

Research Scholar, Bhabha Atomic Research Centre (BARC), Mumbai

Sep 2019 – Present

- Developed a 2D3v electrostatic particle-in-cell (PIC) code in C++ for ion-extraction systems.
- Designed a Poisson's Solver with a novel algorithm using Chebyshev acceleration factor.
- Built a software model for ion transport using fluid and kinetic descriptions.
- Parallelized PIC code using MPI, scaled on Aditya Supercomputing Facility, BARC.

Assistant Professor, Atharva College of Engineering, Mumbai

July 2017 – Sep 2019

- Developed MATLAB scripts and Simulink models to optimize power loss in IEEE bus systems.
- Computed models for IEEE-16, IEEE-33, and IEEE-69 bus systems using Visual Studio.

Assistant Professor, VJTI, Mumbai

Dec 2015 – May 2017

- Worked on power system optimization using MATLAB, Scilab, and Octave.
- Conducted computer-aided power system analysis and state-space analysis.

- Managed billing and maintenance of DL and NDL connections.
- Prepared estimates for new connections and revenue reports.

Technical Projects

Adaptive Meshing Algorithm in C++

Developed an adaptive meshing algorithm in C++ to optimize the computational efficiency of simulations. The algorithm dynamically adjusts mesh density based on localized error estimates, providing higher resolution where needed and reducing computational load in less critical regions. This approach improved simulation accuracy while significantly decreasing runtime. The code was implemented in Visual Studio and is compatible with other simulation modules developed in C++.

Particle-in-Cell Simulation in C++ with High Performance Computing (HPC)

Engineered a 2D3v particle-in-cell (PIC) simulation code in C++ to model electrostatic fields with particle dynamics. The PIC code incorporated efficient data structures and parallel algorithms and was deployed on the Aditya Supercomputing Facility at BARC using MPI for parallel processing. Leveraged HPC to scale the code across multiple processors, achieving substantial computational speedup. This enabled simulations of larger and more complex systems with high precision, providing valuable insights for scientific research.

Ion Extraction Model from Photoplasma in C++

At BARC, developed a comprehensive computational model to simulate ion extraction from photoplasma using various electrode configurations, crucial for optimizing ion beam production in experimental settings. This model was implemented entirely in C++ and integrated advanced parallel computing techniques through MPI, allowing simulations to run efficiently on high-performance computing (HPC) systems.

The model utilized finite element analysis (FEA) and complex numerical methods to calculate electric fields and simulate ion behavior accurately under diverse electrode configurations. The simulation output was visualized with GNU plot, enabling detailed analysis of ion trajectories and density distributions. The adaptive meshing approach was applied to ensure high resolution in regions of interest, which minimized computation time without compromising accuracy. This project demonstrated the effective combination of C++, MPI, FEA, and GNU plot for solving complex physical models with high computational demands.

Poisson Solver with Chebyshev Acceleration in C++

Designed a Poisson Solver in C++ utilizing Gauss-Seidel relaxation with a Chebyshev acceleration factor. The custom solver achieved computational efficiency gains by reducing the number of iterations required for convergence. This solver is integrated into larger simulation models, supporting various research projects at BARC.

Power Loss Optimization for IEEE Bus Systems using MATLAB and C++

Developed MATLAB scripts and Simulink models to perform power flow analysis and optimize power losses in IEEE bus systems (IEEE-16, IEEE-33, and IEEE-69). The MATLAB scripts were designed to perform load flow calculations and identify areas with the highest power loss, facilitating targeted optimizations. Using Simulink, I modeled the dynamic behavior of power systems, incorporating various load and line parameters to simulate real-world conditions accurately.

Additionally, I computed power system models in C++ within Visual Studio, leveraging custom algorithms to enhance computational efficiency for large-scale system simulations. For each IEEE bus system, I implemented methods to balance loads and minimize losses by adjusting configurations and re-routing power flows. The combination of MATLAB, Simulink, and C++ resulted in a highly efficient optimization model that was tested on multiple IEEE bus standards to validate accuracy and scalability.

Publications

Journal Publications

- P. Singh, and N. Maiti, "Investigation on Ion Extraction Processes in M-Type Electrode Configuration and the Influence of Top Electrode", *Physica Scripta*, 99(2), p.025609, 2024. (Impact factor: 3.01).
- P. Singh, G. Sridhar, and N. Maiti, "Modeling and Simulation of PGGP Electrode Configuration for Ion-Extraction Process Using 2.5D-PIC Code," *IEEE Transactions on Plasma Science*, vol. 51, no. 1, pp. 35-48, 2023. (Impact factor: 1.5).
- P. Singh, G. Sridhar, and N. Maiti, "Two-dimensional PIC simulation of the ion-extraction process in an electrostatic field using different electrode configurations." *Journal of Physics: Conference Series*, vol. 2443, 2023. (Impact factor: 0.51).

- P. Singh, G. Sridhar and N. Maiti, "Expansion of a finite-size photoplasma in an electrostatic field with M-type electrode configuration using a novel adaptive mesh 2D-PIC code," *Current Applied Physics*, vol. 47, 2023, pp. 30-43. (Impact factor: 2.8).
- P. Singh, A. Majumder, A. K. Pulhani, G. Sridhar and N. Maiti, "Development of a 1-D Electrostatic Particle-in-Cell Code", *IEEE Transactions on Plasma Science*, vol. 49, no. 10, pp. 3257-3267, 2021. (Impact factor: 1.5).
- Verma, Harish Kumar, and Priti Singh. "Optimal reconfiguration of distribution network using modified culture algorithm." *Journal of The Institution of Engineers (India): Series B* 99 (2018).
- Majumder, A., A. K. Pulhani, A. Ghosh, P. Singh, and N. Maiti. "Need for enrichment of lutetium isotope and design of a laser based separator module." *Applied Radiation and Isotopes* 202 (2023).

Conference Papers

- P. Singh, G. Sridhar and N. Maiti, "Two-dimensional PIC simulation of ion-extraction process in an electrostatic field using different electrode configurations", *14th International Conference on Electron Beam Technology (EBT-2022)*, Varna, Bulgaria.
- P. Singh, G. Sridhar and N. Maiti, "Performance Improvement of a 1DESPIC Particle-in-Cell Code using Adaptive Mesh", *14th International Conference on Plasma Science and Applications (ICPSA-2021)*, Nanjing, China.
- P. Singh, G. Sridhar and N. Maiti, "A Two Dimensional Particle-In-Cell Simulation Code for Pulsed Plasma in an Electrostatic Field", *13th International Conference on Plasma Science and Applications (ICPSA-2020)*, Cuttack, India.
- P. Singh, A. Majumder, A. K. Pulhani, G. Sridhar and N. Maiti, "Methods of High-Density Plasma Generation," *International Conference on Convergence to Digital World - Quo Vadis (ICCDW)*, Mumbai, India.

Awards and Achievements

- Awarded DDFS fellowship by DAE for PhD research (2019-2024).
- Best paper award at IEEE International Conference (ICCDW 2020).
- GATE qualified with 97 percentile (2013-14).

Languages

English, Hindi

Personal Information

- **Date of Birth:** 31st March 1989
- **Address:** 2406, Dosti Oak, Dosti West County, Thane West, Mumbai - 400608